Lecture 22 Notes: Cost of Capital (WACC) and Corporate Valuation

Corporate Finance

I. The Cost of Debt and the Tax Shield

Key Principle

Debt is **tax-deductible**, which provides a **tax shield**. This makes the actual cost of debt lower for the firm.

The Tax Shield Concept

When a firm pays interest on debt, this interest expense is deducted from taxable income, reducing the tax burden.

$$\text{Tax Saving} = \text{Interest Expense} \times \text{Tax Rate}$$

Example

$$\begin{aligned}
\text{Loan Amount} &= \$100,000 \\
\text{Interest Rate} &= 8\% \\
\text{Interest Expense} &= \$100,000 \times 0.08 = \$8,000 \\
\text{Tax Rate}(T) &= 40\% = 0.40 \\
\text{Tax Saving} &= \$8,000 \times 0.40 = \$3,200 \\
\text{After-Tax Interest Cost} &= \$8,000 - \$3,200 = \$4,800
\end{aligned}$$

After-Tax Cost of Debt Formula

$$\boxed{r_d(1-T)}$$

Where:

- r_d = Before-tax cost of debt (interest rate)
- T = Marginal tax rate
- $r_d(1-T)$ = After-tax cost of debt

Important Notes

- Firms take debt because they need funds, not specifically to avoid taxes.
- As a firm's **debt burden increases**, borrowing costs increase due to the **cost of financial distress**.
- We always use **future, anticipated costs** (rates for new loans or new stock issues), not existing or past rates.

II. Methods for Calculating the Cost of Debt

Source of Debt	Cost of Debt Calculation
Bank Loan	Stated rate (e.g., KIBOR + 3%)
Bonds	**Yield to Maturity (YTM)**, not the coupon rate

$$\text{After-Tax Cost of Debt} = \text{YTM} \times (1 - T)$$

III. Cost of Preferred Stock

Cost of Preferred Stock Formula

$$\boxed{r_p = \frac{D_p}{P_p}}$$

Where:

- r_p = Cost of preferred stock
- D_p = Preferred dividend (fixed)
- P_p = Current price of preferred stock

Key Characteristics

- Preferred stock has a **fixed dividend** (similar to bonds but no maturity).
- No tax adjustment because preferred dividends are **not tax-deductible**.

IV. Cost of Common Equity

There are two methods to calculate the cost of common equity:

Method 1: CAPM (Capital Asset Pricing Model)

$$\boxed{r_s = r_{RF} + \beta \times (r_m - r_{RF})}$$

Method 2: DDM (Dividend Discount Model)

$$\boxed{r_s = \frac{D_1}{P_0} + g}$$

Where:

- D_1/P_0 = Dividend yield
- g = Growth rate
- r_s = Cost of common equity

Key Insight

The return an investor requires to invest in a security is exactly what it costs the firm to obtain those funds.

V. Weighted Average Cost of Capital (WACC)

WACC Formula

$$\boxed{WACC = w_d \times r_d(1 - T) + w_p \times r_p + w_s \times r_s}$$

Where:

- w_d = Weight of debt
- w_p = Weight of preferred stock
- w_s = Weight of common equity
- $w_d + w_p + w_s = 1$

Determining Weights

- Weights should represent the firm's **target capital structure**.
- If weights are not given, calculate them based on the **proportions** of each source relative to total capital.

Example: Calculating Weights

$$\begin{aligned}
\text{Debt} &= \$500,000 \\
\text{Preferred Stock} &= \$200,000 \\
\text{Common Equity} &= \$300,000 \\
\text{Total Capital} &= \$1,000,000 \\
w_d &= 500,000/1,000,000 = 0.50 = 50\% \\
w_p &= 200,000/1,000,000 = 0.20 = 20\% \\
w_s &= 300,000/1,000,000 = 0.30 = 30\%
\end{aligned}$$

VI. Flotation Costs

Definition

Flotation costs are costs incurred when a company issues new securities (IPOs, seasoned offerings), including:

- Prospectuses and documentation
- Legal and registration fees
- Investment banker fees

Adjustment for Flotation Costs

$$\text{Net Price Received} = P_0 \times (1 - f)$$

Where:

- P_0 = Market price of the security
- f = Flotation cost as a percentage of the price

Adjusted Cost of Equity with Flotation Costs

$$r_s = \frac{D_1}{P_0(1 - f)} + g$$

VII. The Corporate Valuation Model (Free Cash Flows)

Why Use Corporate Valuation?

For startups or growth stocks that don't pay dividends, the Dividend Discount Model cannot be used. Instead, we use the **Corporate Valuation Model** based on **Free Cash Flows (FCF)**.

Definition of Free Cash Flow (FCF)

FCF is the cash a firm generates that is actually **available for distribution** to all capital providers (debt holders and equity holders) after all operating expenses and investments.

FCF Formula

$$\boxed{FCF = EBIT(1 - T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC}}$$

Where:

- $EBIT(1 - T)$ = After-tax operating income (because EBIT is before interest)
- Depreciation = Added back because it is a **non-cash charge**
- CapEx = Capital Expenditures (investment in fixed assets)

- ΔNOWC = Change in Net Operating Working Capital

Key Insight

We use **EBIT (not net income)** because FCF is for **everyone** (both debt holders and equity holders). We do not subtract interest yet.

VIII. Determining Intrinsic Value Using FCF

Step 1: Value the Entire Firm

Project future FCFs and discount them to the present using **WACC** as the discount rate.

$$\boxed{V_{Firm} = \sum_{t=1}^{\infty} \frac{FCF_t}{(1 + WACC)^t}}$$

Constant Growth FCF Model

If FCF grows at a constant rate (g):

$$\boxed{V_{Firm} = \frac{FCF_1}{WACC - g}}$$

Where $FCF_1 = FCF_0 \times (1 + g)$ and $g < WACC$.

Step 2: Calculate the Value of Equity

$$\boxed{V_{Equity} = V_{Firm} - V_{Debt} - V_{Preferred}}$$

Step 3: Calculate Intrinsic Value per Share

$$\boxed{\hat{P}_0 = \frac{V_{Equity}}{\text{Number of Shares Outstanding}}}$$

IX. Market Efficiency and Investment Decisions

Market Efficiency

In a competitive market:

- When an undervalued stock is identified, investors start buying it.
- This causes the price to rise until it equals the intrinsic value.
- Professional analysts publish "target prices" in research reports.

Decision Rule

Condition	Action
$\hat{P}_0 > P_0$ (Undervalued)	**BUY** (price should rise)
$\hat{P}_0 < P_0$ (Overvalued)	**SELL** (price should fall)

X. Summary: Cost of Capital Components

Component	Formula	Tax Adjustment?
Debt	$r_d(1-T)$	Yes (tax-deductible)
Preferred Stock	D_p/P_p	No
Common Equity (CAPM)	$r_{RF} + \beta(r_m - r_{RF})$	No
Common Equity (DDM)	$D_1/P_0 + g$	No
WACC	$w_d r_d(1-T) + w_p r_p + w_s r_s$	N/A

XI. Summary: Corporate Valuation Model Steps

1. Calculate **Free Cash Flows (FCF)** using:

$$FCF = EBIT(1-T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC}$$

2. Calculate the **value of the firm** (V_{Firm}) by discounting FCFs at WACC.
3. Calculate the **value of equity**:

$$V_{Equity} = V_{Firm} - V_{Debt} - V_{Preferred}$$

4. Calculate the **intrinsic value per share**:

$$\hat{P}_0 = \frac{V_{Equity}}{\text{Shares Outstanding}}$$

5. Compare $\hat{P}_0$ to the market price to make investment decisions.

Formula Summary Sheet

Key Formulas

$$\text{After-Tax Cost of Debt}: \quad r_d(1-T) \tag{1}$$

$$\text{Cost of Preferred Stock}: \quad r_p = \frac{D_p}{P_p} \tag{2}$$

$$\text{Cost of Equity (CAPM)}: \quad r_s = r_{RF} + \beta(r_m - r_{RF}) \tag{3}$$

$$\text{Cost of Equity (DDM)}: \quad r_s = \frac{D_1}{P_0} + g \tag{4}$$

$$\text{WACC}: \quad WACC = w_d r_d(1-T) + w_p r_p + w_s r_s \tag{5}$$

$$\text{Free Cash Flow}: \quad FCF = EBIT(1-T) + \text{Depreciation} - \text{CapEx} - \Delta\text{NOWC} \tag{6}$$

$$\text{Firm Value (Constant Growth)}: \quad V_{Firm} = \frac{FCF_1}{WACC - g} \tag{7}$$

$$\text{Equity Value}: \quad V_{Equity} = V_{Firm} - V_{Debt} - V_{Preferred} \tag{8}$$

$$\text{Intrinsic Value per Share}: \quad \hat{P}_0 = \frac{V_{Equity}}{\text{Shares Outstanding}} \tag{9}$$